# Workshop Setup 🚀

Completing this setup before the workshop ensures we can dive straight into coding without install delays.

## Prerequisites

- **Git:** https://git-scm.com/downloads
- **uv (package manager):** https://docs.astral.sh/uv/getting-started/installation/
  Used to manage project dependencies and ensure consistent Python versions. After installing `uv`, restart the terminal.
- **A code editor/IDE:** For example, VS Code: https://code.visualstudio.com/download
- **A way to work with Jupyter Notebooks:** For example, the Jupyter extension for VS Code: https://code.visualstudio.com/docs/datascience/jupyter-notebooks

## Step 1: Clone the course repo

Run:

```
git clone https://github.com/arilamstein/streamlit-workshop.git
cd streamlit-workshop
```

**github.com/arilamstein/streamlit-workshop**

Please do the setup if you have not done so already.

# Building Data Apps In Python With Streamlit

**Session 1: Intro**

**Ari Lamstein**
AriLamstein.com

## Exercise: What is a Data App?

Type your answer to these 2 questions in the chat:

1. What do you already know about Data Apps, Python and Streamlit?
2. What do you want to build after completing this course?

## About me

**Ari Lamstein**

- Former software engineer in San Francisco (~10 yrs)
- Trainer and open-source contributor (R)
- Transitioned to Python in recent years
- Built civic data apps with Streamlit: *Covid Demographics Explorer* and *Immigration Enforcement Explorer*

Before I give my own answer about what a "data app" is, let me introduce myself

## Exercise: What is a Data App?

Type your answer to these 2 questions in the chat:

1. What do you already know about Data Apps, Python and Streamlit?

Streamlit is really popular: downloaded 16M times last month

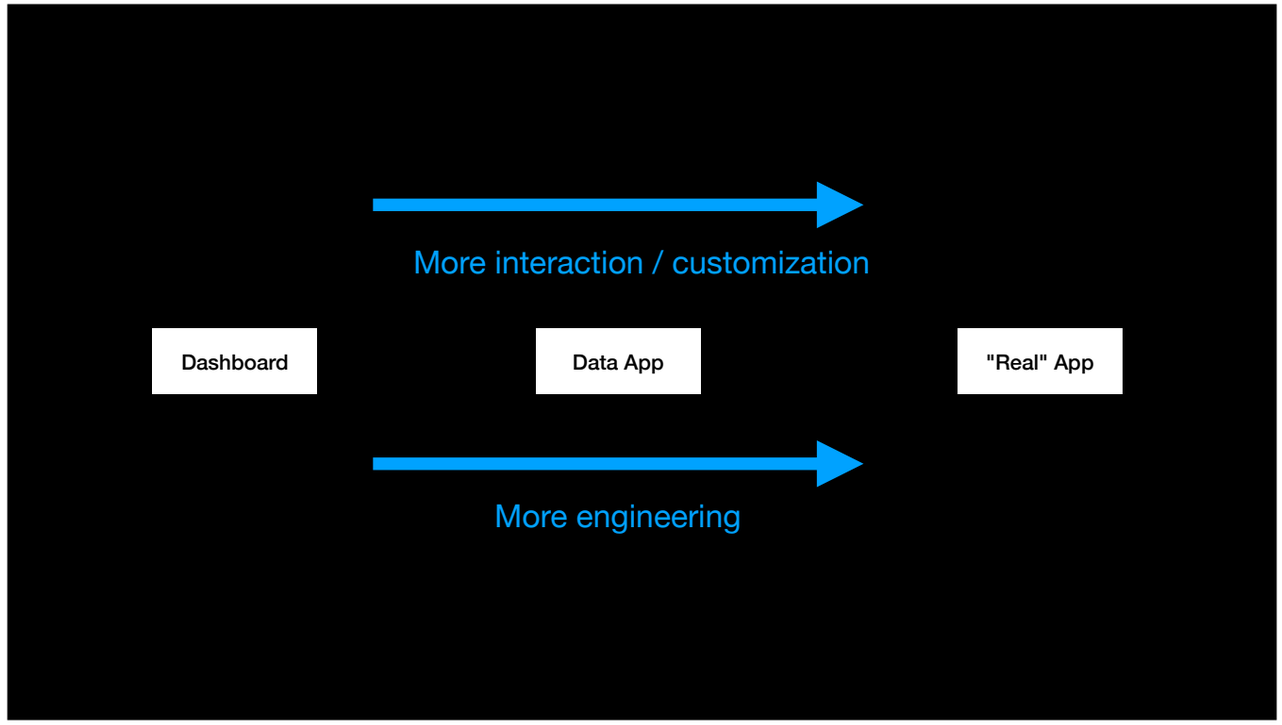
2. What do you want to build after completing this course?

I use Streamlit to build data apps that make it easy to understand public / government datasets. This is the course I wish I had when I was starting out!

This is my "short answer" to the question. On the next slide I'll give a longer question.

## Exercise: What is a Data App?

- My answer has two parts
  - "Data" apps vs."Real" apps vs. Dashboards
  - Data apps vs. Jupyter Notebooks




- A Dashboard has no interactivity whatsoever - think of the dashboard of car. And as a result, there are a lot of ways to create them with using little to no code.
- A "real" app, like Amazon, has a lot of interaction: creating an account, browsing items, purchase items and leave reviews. As a result, there is a lot of engineering involved.
- Data Apps are somewhere in the middle. Like a dashboard, they're designed to inform. But unlike a dashboard, you can interact with them. But unlike a "Real" app, that interaction is typically limited to exploring a dataset.
- Data apps require *some* engineering. In this course, we'll be writing small amounts of Python code. Streamlit will take care of all the front end code (like HTML, CSS and JavaScript). We won't be using a database / writing SQL.

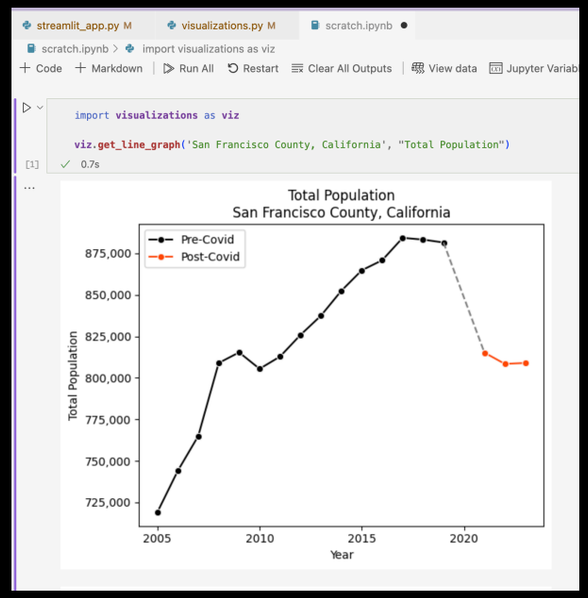
## Data Apps vs. Jupyter Notebooks

- Jupyter Notebooks let me create a single story about a dataset. I can share the notebook with others for free via github.
- Streamlit lets me create an app to explore a dataset. I can share the app with others for free via Streamlit Cloud.
- I often start with a notebook, and end with a Streamlit app.

Ask: How many people here have used Jupyter notebooks before?
In my own work, I view Streamlit as a companion to Jupyter notebooks.

## Example Jupyter Notebook

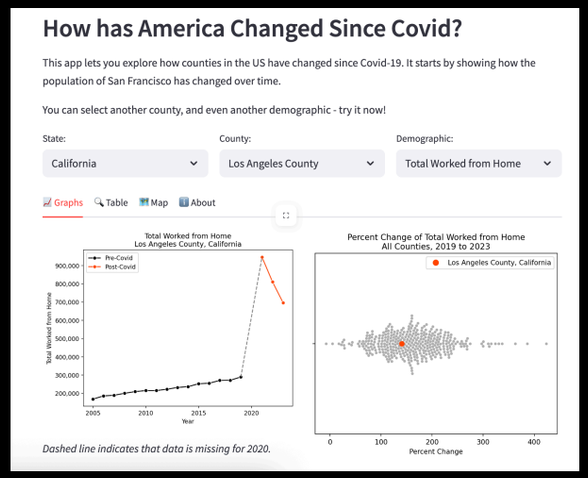
- This notebook tells a story about one city.




At one point I was interested in understanding how San Francisco's population changed during Covid. My first step was to create a Jupyter Notebook that had graphs like this.

# Example Streamlit App

- This Streamlit App lets users see how the population of their county changed
- https://census-explorer.streamlit.app/
- You'll be creating a similar app today




Note the interactivity:

-select boxes -> graphs change.

-layout: tabs are clickable

The app you'll be creating today is similar to this. But it will focus on states instead of counties.

## Learning Goals: Intro

1. Exercise: "Rerun" a Streamlit App
2. Exercise: Running a Jupyter Notebook

The goal in this introductory module is to make sure that you are able to complete the rest of the course. To that end, we're now going to do 2 exercises: the first is to learn how to "rerun" a streamlit app.

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cd streamlit-workshop
```

**github.com/arilamstein/streamlit-workshop**

I started the workshop by asking everyone if they had done the setup. From this point on, I'm going to assume that everyone has at least cloned the repo, installed uv and created the virtual environment. Does anyone need help with that?

## Activate the Virtual Environment

- Activate the virtual environment:
  - **Mac/Linux**: source .venv/bin/activate
  - **Windows (Command Prompt)**: .venv\Scripts\activate
  - **Windows (PowerShell):** .venv\Scripts\Activate.ps1

- Test that the virtual environment is activated:
  - Run python --version → should report a version starting with **Python 3.13**
  - Run which python (Mac/Linux) or where python (Windows) → should show a path inside **.venv**

The command to activate the virtual environment depends on your operating system. Type the right command now

## Run the Demo App

- Each session has its own directory
- Type `cd 1-intro-setup`
  - This moves to the directory for this session
- Type `streamlit run streamlit_app.py`
  - This runs the demo app for this section

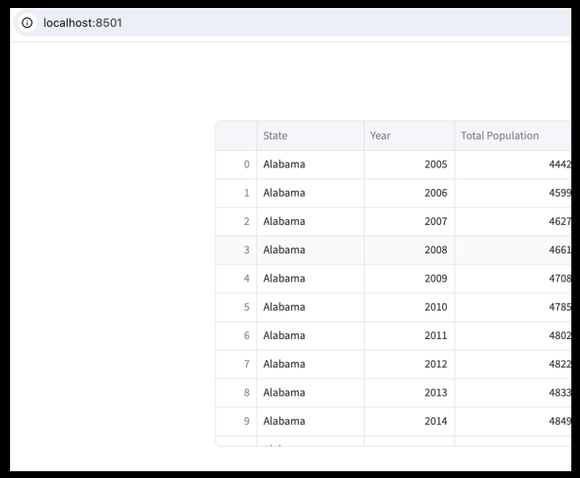
```
(streamlit-workshop) ➜  streamlit-workshop git:(main) ✗ cd 1-intro-setup
(streamlit-workshop) ➜  1-intro-setup git:(main) ✗ streamlit run streamlit_app.py

  You can now view your Streamlit app in your browser.
```

Move to the "1-intro-setup" directory and type "streamlit run streamlit_app.py"

# Running the App

- It should open a browser like this
- Click the table headers to sort (e.g. State, Year, Total Population)
- **Quiz:**
  - What are the min and max years of the dataset?
  - What state had the lowest population?
  - What state had the highest income?




| | State | Year | Total Population |
|---|---|---|---|
| 0 | Alabama | 2005 | 4442 |
| 1 | Alabama | 2006 | 4599 |
| 2 | Alabama | 2007 | 4627 |
| 3 | Alabama | 2008 | 4661 |
| 4 | Alabama | 2009 | 4708 |
| 5 | Alabama | 2010 | 4785 |
| 6 | Alabama | 2011 | 4802 |
| 7 | Alabama | 2012 | 4822 |
| 8 | Alabama | 2013 | 4833 |
| 9 | Alabama | 2014 | 4849 |

# How the App Works

- Open the file streamlit_app.py in the directory 1-intro
- Two imports:
  - pandas as pd
  - streamlit as st

session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

#st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```

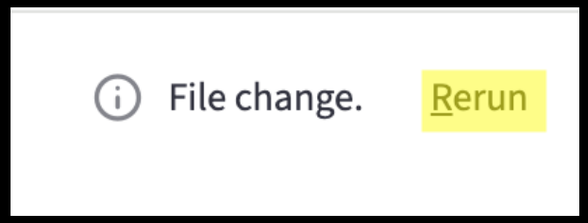
Until now I've just asked you to make sure you can run the app on your machine. Now let's take a look at how it works. It's really just 4 lines of code - and two are import statements. This simplicity is really impressive!

I assume most people have at least some exposure to Pandas - here we use it to read in a csv file as a dataframe. And we store the result in a variable called df. Pandas is normally imported with the alias pd.

Streamlit is normally imported with the alias st. So all the functions you learn today will start with "st dot". Here we use the function st.dataframe to make the app print a dataframe with sortable columns.

## Rerun

- Streamlit is designed for fast iteration
- Each time you modify and save the script the word Rerun will appear in the top right of the app
- Click it to rerun the app with your changes
- Do not stop and restart the app from the command line




In the next exercise I'm going to ask you to make a small change to the app and then rerun it.

# Exercise: Rerun

- To start:
  - Open the file streamlit_app.py in the directory 1-intro
  - Type streamlit run streamlit_app.py to run the app
- Then:
  - Uncomment the st.title line
  - Save the file
  - Click "rerun" and verify that the title appears

session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

#st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```

This is an exercise to make sure you're comfortable changing the app, and clicking "rerun" to see your changes.

# Result

## Demo Streamlit App

| | State | Year | Total Population | Median Household Income |
|---|---|---|---|---|
| 0 | Alabama | 2005 | 4442558 | 36879 |
| 1 | Alabama | 2006 | 4599030 | 38783 |
| 2 | Alabama | 2007 | 4627851 | 40554 |
| 3 | Alabama | 2008 | 4661900 | 42666 |
| 4 | Alabama | 2009 | 4708708 | 40489 |
| 5 | Alabama | 2010 | 4785298 | 40474 |
| 6 | Alabama | 2011 | 4802740 | 41415 |
| 7 | Alabama | 2012 | 4822023 | 41574 |
| 8 | Alabama | 2013 | 4833722 | 42849 |
| 9 | Alabama | 2014 | 4849377 | 42830 |

# Streamlit

- Lets the script run as a webapp
- Cannot run in a Jupyter notebook - must be a regular .py file.
- Imported as st
- Two output functions:
  - st.title
  - st.dataframe
- Processed top to bottom.

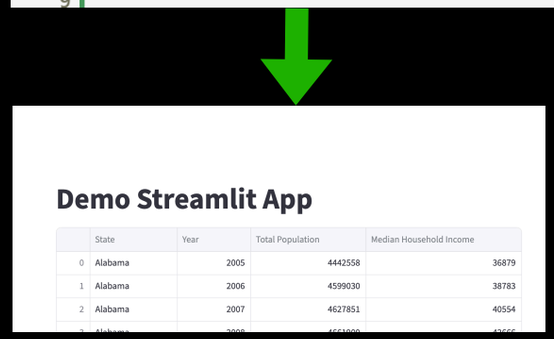
session1 > streamlit_app.py > ...

```python
import streamlit as st
import pandas as pd

st.title("Demo Streamlit App")

df = pd.read_csv("state_data.csv")

st.dataframe(df)
```




Our demo app has the line "import streamlit as st".

# The data

- state_data.csv
- Basic demographic data of US States
- Comes from the American Community Survey (ACS)
- Each state, 2005-2023

session1 > state_data.csv > data

```csv
State,Year,Total Population,Median Household Income
Alabama,2005,4442558,36879
Alabama,2006,4599030,38783
Alabama,2007,4627851,40554
Alabama,2008,4661900,42666
Alabama,2009,4708708,40489
Alabama,2010,4785298,40474
Alabama,2011,4802740,41415
Alabama,2012,4822023,41574
Alabama,2013,4833722,42849
Alabama,2014,4849377,42830
Alabama,2015,4858979,44765
Alabama,2016,4863300,46257
```

## Learning Goals: Intro

1. Exercise: "Rerun" a Streamlit App
2. Exercise: Running a Jupyter Notebook

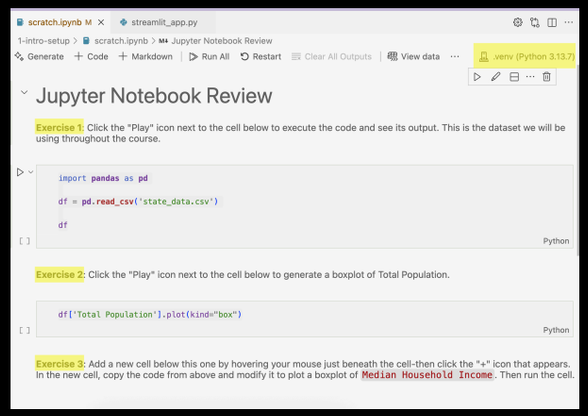
We're going to be using Jupyter Notebooks throughout this course.

# Exercise: Jupyter Notebooks

- Open review.ipynb. Two ways:
  1. VS Code with Jupyter plugin
  2. Jupyter notebook via command line:

     uv run jupyter notebook scratch.ipynb

- Complete the 3 exercises as a review




Later on we'll be using Jupyter Notebooks. I want to include with notebooks now to make sure everyone is comfortable with them. Note that in VS Code you'll need to select a "kernel" in the upper right hand corner.

## Section 1 Review

- What is a data app?
- Confirm setup:
  - Clone course repo
  - Create virtual environment with uv
  - Activate virtual environment
- 2 Exercises:
  - Modify & Rerun app
  - Jupyter Notebooks